MapSORT: Landmark Association for Identity-Persistent Tracking of Stationary Crops and Fruit

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Abstract

Purpose: Plant- and fruit-level records in precision agriculture, such as yield estimates, phenotyping logs and targeted spraying, require every plant or fruit to keep one identity across a video, including when the camera leaves and later returns to it. General-purpose multi-object trackers lose identity in exactly this situation, because crops look alike and an object out of view for more than a few frames is forgotten. This work turns the property that sets agricultural targets apart, that they do not move, into an association cue that keeps identities intact across occlusions and revisits.

Methods: MapSORT reconstructs the scene with structure-from-motion and gives every detection and every track a landmark: a Gaussian over the object's position in the scene map. Built on BoT-SORT, it fits a camera to its own tracks in every frame, from their landmarks and boxes, and uses that camera to place each stationary object in the image. Tracks the frame-to-frame association misses are re-matched where their landmark projects; lost tracks, and tracks that left the view long ago and were kept dormant rather than deleted, are revived the same way, including objects that re-enter at the frame edge. MapSORT is compared with StrongSORT, Deep OC-SORT and BoT-SORT on four public datasets (LettuceMOTS, MOrangeT, GrapeMOTS, AppleMOTS).

Results: Averaged over the four datasets, MapSORT reaches 95.5 HOTA and 96.0 IDF1, 10.2 and 16.7 points above the strongest baseline, BoT-SORT, and cuts identity switches from 1150, the fewest of any baseline, to 223. It preserves identity in 91.9% of 1043 revisit events, against at most 11.5% for the baselines, at every length of absence. It has fewer identity switches than every baseline on 19 of the 21 test sequences.

Conclusion: The position of a stationary object in a map of the scene is an identity cue that survives arbitrarily long absences. Recovering the camera from the tracks themselves turns that position into a motion model and a re-identification cue, giving identity-consistent tracks and accurate counts across crops, platforms and viewing patterns.

Keywords: Multi-object tracking, Structure-from-motion, Camera resection, Fruit counting, Plant re-identification, Yield estimation

1 Introduction

Many precision-agriculture tasks operate at the level of the individual plant or fruit. Yield estimation from video counts fruit by counting tracks (Stein et al, 2016; Liu et al, 2019; Häni et al, 2020; Villacrés et al, 2023). Per-plant phenotyping attaches measurements from successive frames and passes to the same plant. Selective spraying and harvesting require an agricultural robot to know which plant it is looking at. All of these depend on multi-object tracking (MOT) that assigns each physical object one identity, and keeps it, for as long as the object is in the video.

Agricultural video makes this hard in a specific way. Plants in a row, oranges on a tree and grape bunches on a cane are visually near-identical, so appearance embeddings learned for pedestrians separate them poorly. Leaves, branches and neighbouring fruit occlude them. Most importantly, the camera routinely leaves an object and comes back to it: a ground robot drives a row and returns along it, a handheld camera sweeps the canopy more than once, a UAV inspects a vine from several perspectives. Tracking-by-detection methods (Bewley et al, 2016; Wojke et al, 2017; Zhang et al, 2022; Aharon et al,

2022; Du et al, 2023) predict object motion in the image plane and keep an unmatched track alive for a fixed number of frames. An object that is out of view for longer is deleted and receives a new identity when it reappears. Every such switch inflates the count and fragments the plant's record (Fig. 1).

The targets of agricultural tracking have a property that pedestrians and vehicles do not: they are stationary. A lettuce occupies the same place in the field on the outbound and return pass; an orange hangs at the same point of the canopy however often the camera sweeps past it. Its position in a three-dimensional map of the scene is therefore an identity cue that does not decay with time out of view and does not depend on appearance. Structure-from-motion (SfM) recovers such a map from the same monocular RGB video the tracker consumes (Schönberger and Frahm, 2016).

A stationary object's map position is only useful for tracking if it can be compared with what the camera sees now. Comparing positions in 3D, as earlier fruit trackers do, is fragile: a landmark triangulated from a few keypoints is uncertain mostly along the viewing ray, which is exactly the direction that separates neighbouring fruit in depth. Projecting the landmark into the current image removes that direction, but needs the current camera. This paper presents MapSORT, which obtains that camera from the tracker itself. Every frame, the tracks that a standard frame-to-frame tracker has just matched supply correspondences between their landmarks and their boxes, and a robust resection recovers the frame's camera from them. With the camera, every stationary object, tracked, lost or long gone, has a predicted position in the current image. MapSORT uses it twice: as a motion model that re-matches tracks the frame-to-frame association missed, and as a re-identification cue that revives a track when its object comes back into view, however long it was away.

The contributions of this work are:

1. A per-frame camera resection from the tracker's own landmark-to-box correspondences, robust to wrong matches and to near-coplanar scenes such as a field of lettuce, which places every stationary object, visible or not, in the current image.

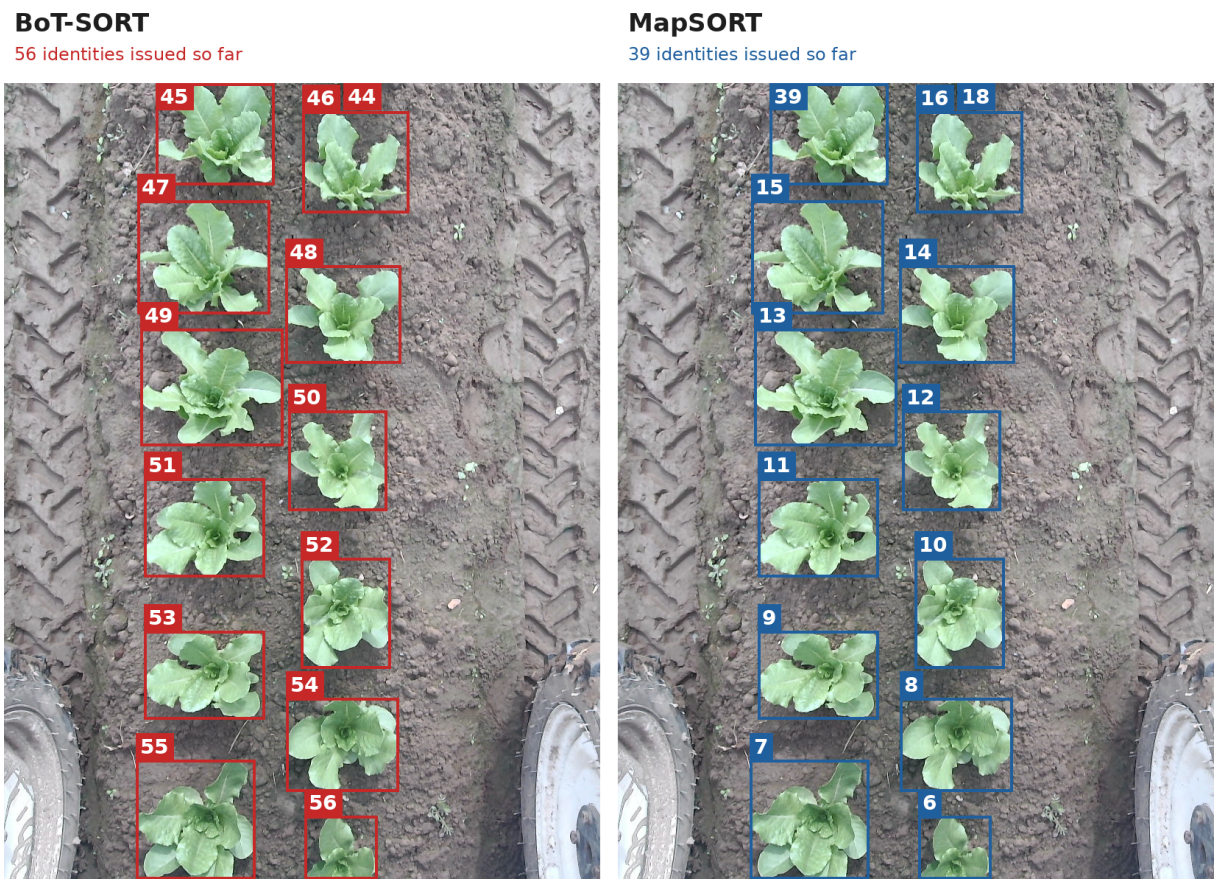


Fig. 1 The same frame of a field robot's return pass along two lettuce rows (LettuceMOTS 0011), tracked by BoT-SORT (left) and MapSORT (right). Identities are numbered in order of first appearance. BoT-SORT has forgotten the plants it saw on the way out and has issued 56 identities for 37 plants; every plant in view carries a new one. MapSORT gives each plant the identity it received on the outbound pass and has issued 39.

2. MapSORT, which adds two stages to BoT-SORT (Aharon et al, 2022): a map-motion stage that re-matches unmatched tracks where the camera places them, and a revival stage that reclaims the identity of lost and dormant tracks by reprojection, including objects that re-enter at the frame edge with only a sliver visible. Tracks that leave the view are kept dormant instead of deleted, so identity survives absences of any length.
3. A revisit-centred evaluation protocol: a revisit identity preservation rate that decomposes CLEAR identity switches by the absence that preceded them, together with count error, which is the quantity yield estimation consumes.
4. A benchmark against StrongSORT, Deep OC-SORT and BoT-SORT on four public agricultural datasets (LettuceMOTS, MOrangeT, GrapeMOTS and AppleMOTS), covering four crops, four platforms and three viewing patterns, with per-sequence results.

2 Related work

2.1 Tracking-by-detection

SORT (Bewley et al, 2016) established the tracking-by-detection template: a constant-velocity Kalman filter (Kalman, 1960) predicts each track's box, and Hungarian matching (Kuhn, 1955) on intersection-over-union (IoU) assigns detections to tracks. DeepSORT (Wojke et al, 2017) added an appearance embedding and a matching cascade that prioritises recently seen tracks. ByteTrack (Zhang et al, 2022) recovers low-confidence detections in a second association round; OC-SORT (Cao et al, 2023) corrects Kalman drift during occlusion from observations, and Deep OC-SORT (Maggiolino et al, 2023) adds camera-motion compensation and an appearance cost whose weight adapts to how distinctive each detection's embedding is; FairMOT (Zhang et al, 2021) learns detection and embedding jointly. StrongSORT (Du et al, 2023) modernises DeepSORT with a stronger re-identification backbone and fused appearance and motion costs. BoT-SORT (Aharon et al, 2022) compensates camera motion with a global image warp, refines the Kalman state, and fuses IoU and appearance costs; it is the base of MapSORT. SAM2MOT (Jiang et al, 2025) departs from box association altogether: it uses the SAM 2 video segmentation model (Ravi et al, 2024) to propagate object masks through time, so identity follows from segmentation rather than from matching.

All of these methods reason about identity in the image plane or in an appearance space. Neither is stable when the camera leaves an object and returns: image-plane motion over a long absence is not predictable, a global warp cannot follow the parallax of a three-dimensional canopy, and appearance galleries are bounded in both size and age. MapSORT keeps BoT-SORT's appearance and motion machinery and adds a cue that remains valid over any absence.

2.2 Tracking and counting in crops and orchards

Video-based fruit counting tracks fruit to avoid counting it more than once. Halstead et al (2018) tracked sweet pepper with IoU association on a robotic platform. Häni et al (2020) compared detection and counting pipelines for apple yield mapping, and Villacrés et al (2023) compared tracking-by-detection algorithms for apple production estimation. Hu et al (2022) released LettuceMOT with back-and-forth and out-and-in robot trajectories and showed that re-occurring plants defeat standard trackers; the same group introduced LettuceMOTS with a shape-feature MOTS method for precision spraying (Hu et al, 2024). Datasets for homogeneous-object MOTS exist for apples (de Jong et al, 2022) and grape bunches captured from multiple UAV perspectives (Ariza-Sentís et al, 2024), following the MOTS task definition of Voigtlaender et al (2019).

2.3 Three-dimensional localisation for fruit identity

Several works use multi-view geometry to localise fruit. Stein et al (2016) triangulated mango detections from a GNSS/INS-referenced vehicle to count fruit per tree. Liu et al (2019) combined monocular SfM with semantic data association to count and map fruit. Gené-Mola et al (2020) projected instance segmentations onto an SfM point cloud to locate apples in 3D. Rapado-Rincón et al (2023) tracked tomatoes in 3D from a robot arm with known camera poses. Santos et al (2024) relocalised oranges in 3D to recover them after occlusion and showed that removing relocalisation more than halves HOTA on MOrangeT.

These approaches either compare fruit positions in 3D with a fixed, hand-tuned distance threshold, or rely on camera poses from the reconstruction or an external sensor. MapSORT differs in three ways. Its landmarks are Gaussians whose uncertainty shrinks as observations are fused, and its 3D gate is

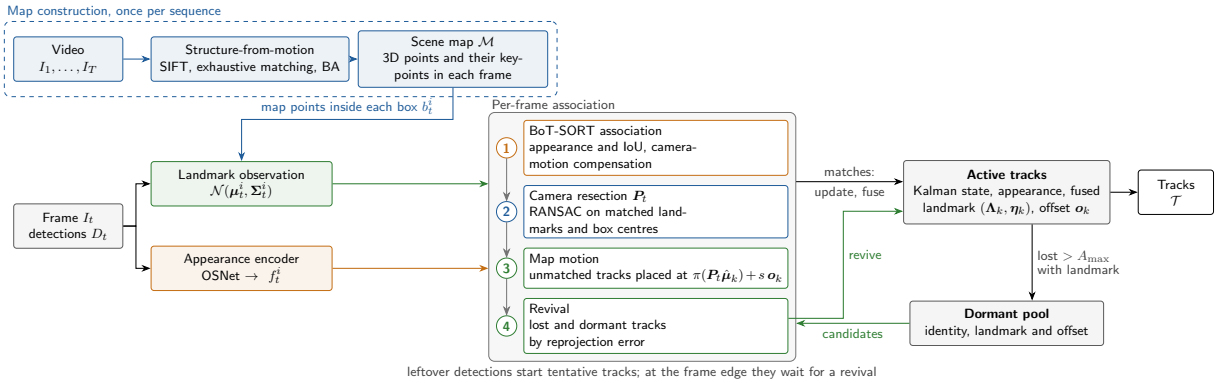


Fig. 2 MapSORT. A scene map is reconstructed once per sequence by structure-from-motion. At each frame, every detection gets an appearance embedding and a landmark observation read from the map points it encloses. BoT-SORT's association runs first (1). Its matches, with the tracks' fused landmarks, yield the frame's camera (2), which places every stationary track in the image: tracks still unmatched are re-matched where it places them (3), and lost and dormant tracks are revived where their landmarks project (4). A confirmed track lost for longer than $A_{\max}$ frames becomes dormant instead of being deleted, so a returning object reclaims its identity.

a Mahalanobis distance that does not depend on the arbitrary scale of a monocular reconstruction. It recovers the camera that relates the map to the current image from its own tracks in every frame, so the projection it compares against is consistent with what the tracker currently believes. And it integrates the landmark cue into a modern tracker as a motion model and a revival stage, rather than replacing association.

3 Materials and methods

3.1 Problem formulation

The input is a monocular RGB video $\{I_1, \ldots, I_T\}$ captured by a moving camera, with no depth, odometry or positioning sensor, and at each frame t a set of detections $D_t = \{d_t^1, \ldots, d_t^{m_t}\}$, each an axis-aligned box b_t^i with centre $\boldsymbol{c}_t^i$ and size s_t^i, the geometric mean of its width and height. The scene contains stationary objects. The task is to produce tracks $\mathcal{T} = \{\tau_1, \ldots, \tau_K\}$, $\tau_k = \{(t, d_t^i)\}$, such that each track corresponds to exactly one physical object and each object to exactly one track: identity must be preserved through occlusion, visual ambiguity and camera motion, including when the camera returns to an object after it has been out of view.

3.2 Overview

Fig. 2 shows the MapSORT pipeline. A scene map is reconstructed once per sequence by SfM (Sect. 3.3), and every detection receives a landmark observation read from it (Sect. 3.4). Tracks fuse their observations into a landmark (Sect. 3.5). At every frame, BoT-SORT's own association runs first. The pairs it matches, with the tracks' landmarks, yield the frame's camera (Sect. 3.6). The camera then drives two further stages: tracks left unmatched are tried again where it places them, and lost and dormant tracks are revived where their landmarks project (Sect. 3.7). Tracks that leave the view are kept dormant rather than deleted (Sect. 3.8). Algorithm 1 summarises one frame.

3.3 Scene map from structure-from-motion

The scene map is a sparse SfM reconstruction of the sequence computed with COLMAP (Schönberger and Frahm, 2016). SIFT features (Lowe, 2004) are extracted at full resolution, because the finest scale-space octaves carry the small objects, fruit and bunches, from which landmarks are read. All frames share one camera with a radial distortion model. Matching is exhaustive rather than sequential, so that frames from a revisit are matched directly to frames from the first pass and the two passes share 3D points; this is what places a returning object at the same map position. The two 1080p GrapeMOTS sequences, which have no revisits and up to 927 frames, are instead matched sequentially, each frame against its 30 neighbours. Incremental mapping with bundle adjustment then yields a point cloud and, for every registered frame, the 2D keypoints that observe each 3D point. The frames carry no camera metadata. One sequence per camera is therefore mapped with free intrinsics; where that fragments the model, it is

mapped again with the focal length of its largest fragment held fixed. The resulting camera is held fixed while the other sequences from the same camera are mapped, and refined in a final bundle adjustment. The largest reconstructed model is kept. MOrangeT ships a COLMAP model of every sequence with all frames registered (Santos et al, 2024), which is used as it is. The frame of reference and scale of a model are arbitrary; MapSORT never needs metric scale.

3.4 Landmark observations

For detection d_t^i with box b_t^i, let $\mathcal{P}_t^i = \{\boldsymbol{X}_j\}_{j=1}^n$ be the 3D points whose keypoints in frame t fall inside b_t^i. Three optional filters, chosen per sequence, remove points that belong to the background rather than the object or are poorly triangulated. A mask filter keeps only points whose keypoint falls on the detection's instance mask, optionally eroded by a few pixels where the mask boundary bleeds into the background; when no point falls on the mask, the points in the box are kept. A track-length filter keeps points observed in at least $\ell_{\min}$ frames. A depth filter drops points more than k median absolute deviations deeper than the median camera-frame depth of the box's points, which removes background seen through gaps in the canopy. The landmark observation $z_t^i = (\boldsymbol{\mu}_t^i, \boldsymbol{\Sigma}_t^i, \boldsymbol{S}_t^i)$ is

$$\boldsymbol{\mu}_t^i = \operatorname{median}(\mathcal{P}_t^i), \tag{1}$$

$$\boldsymbol{S}_t^i = \operatorname{Cov}(\mathcal{P}_t^i) + r^2 \boldsymbol{I}, \qquad r = \tfrac{1}{4}\,\|b_t^i\|\,\bar{z}/f, \tag{2}$$

$$\boldsymbol{\Sigma}_t^i = \boldsymbol{S}_t^i / n, \tag{3}$$

where the median is taken per coordinate, $\|b_t^i\|$ is the box diagonal in pixels, $\bar{z}$ the median camera-frame depth of the points and f the focal length. The extent $\boldsymbol{S}_t^i$ is the object's physical spread in the map frame; the isotropic floor $r^2\boldsymbol{I}$, the object's apparent radius back-projected to its depth, keeps it well conditioned when only a few points survive. The covariance $\boldsymbol{\Sigma}_t^i$ is the uncertainty about where the object's *centre* lies, which shrinks as more points support the estimate. Only $\boldsymbol{\Sigma}_t^i$ is fused across observations; the extent is a property of the plant or fruit and does not shrink however often the object is observed. A detection whose box encloses no usable point carries no landmark, $z_t^i = \varnothing$, and takes part only in BoT-SORT's association.

3.5 Landmark fusion

Each track k maintains its landmark in information form, accumulating the precision-weighted observations assigned to it:

$$\boldsymbol{\Lambda}_k = \sum_{(t,i)\in\tau_k} \left(\boldsymbol{\Sigma}_t^i\right)^{-1}, \qquad \boldsymbol{\eta}_k = \sum_{(t,i)\in\tau_k} \left(\boldsymbol{\Sigma}_t^i\right)^{-1} \boldsymbol{\mu}_t^i, \qquad \hat{\boldsymbol{\Sigma}}_k = \boldsymbol{\Lambda}_k^{-1}, \quad \hat{\boldsymbol{\mu}}_k = \boldsymbol{\Lambda}_k^{-1}\boldsymbol{\eta}_k. \tag{4}$$

This is the static-state information filter (Thrun et al, 2005): because the object does not move, there is no process noise, and the fused covariance $\hat{\boldsymbol{\Sigma}}_k$ contracts with every observation. Where MapSORT compares landmarks in 3D, it uses the squared Mahalanobis distance between an observation and a fused landmark under their combined uncertainty,

$$d_{ki}^2 = \left(\boldsymbol{\mu}_t^i - \hat{\boldsymbol{\mu}}_k\right)^\top \left(\hat{\boldsymbol{\Sigma}}_k + \boldsymbol{\Sigma}_t^i\right)^{-1} \left(\boldsymbol{\mu}_t^i - \hat{\boldsymbol{\mu}}_k\right), \tag{5}$$

and forbids pairs with $d_{ki}^2 > \tau$.

Proposition 1. *d_{ki}^2 is invariant to the similarity transform relating a monocular reconstruction to the metric scene.*

Proof. Under $\boldsymbol{X} \mapsto s\boldsymbol{R}\boldsymbol{X} + \boldsymbol{t}$, every median, sample covariance, depth and back-projected radius in Eqs. (1) to (3) transforms covariantly, so $\boldsymbol{\mu} \mapsto s\boldsymbol{R}\boldsymbol{\mu} + \boldsymbol{t}$ and $\boldsymbol{\Sigma} \mapsto s^2\boldsymbol{R}\boldsymbol{\Sigma}\boldsymbol{R}^\top$; Eq. (4) preserves this. The difference in Eq. (5) scales by $s\boldsymbol{R}$ and the inverse covariance by $s^{-2}\boldsymbol{R}\,(\cdot)\,\boldsymbol{R}^\top$, which cancel. □

The gate τ is therefore a single dimensionless constant that transfers across sequences, crops and reconstructions, a property a fixed metric distance threshold lacks.

3.6 Camera resection from tracked landmarks

A landmark built from a few keypoints is uncertain mostly along the viewing rays that triangulated it, which is the direction separating a fruit from its neighbours behind or in front of it. Projected into

the current image, that error largely disappears. MapSORT therefore compares landmarks in the image, through a camera $\boldsymbol{P}_t \in \mathbb{R}^{3\times4}$ fitted in every frame to the tracker's own state.

The correspondences are the pairs $(\hat{\boldsymbol{\mu}}_k, \boldsymbol{c}_t^i)$ of a track's fused landmark and a detection's box centre, taken from two sources: the track-detection pairs BoT-SORT has just matched (Sect. 3.7), and pairs of a track and a detection whose landmarks are mutual nearest neighbours under Eq. (5) within the gate τ. Errors are measured in box sizes, $e = \|\pi(\boldsymbol{P}_t\hat{\boldsymbol{\mu}}_k) - \boldsymbol{c}_t^i\|/s_t^i$ with π the perspective division, so one threshold serves small fruit and large plants alike. $\boldsymbol{P}_t$ is estimated by RANSAC (Fischler and Bolles, 1981), with the previous frame's camera competing against the random hypotheses and the winner refitted on its inliers for as long as that does not lose inliers. A camera is accepted when at least 8 correspondences agree with it to within half a box size.

Two camera models compete. The full projective camera is fitted by the direct linear transform from six correspondences (Hartley and Zisserman, 2004). It is ill-conditioned when the landmarks are close to coplanar, as on a field of lettuce seen from above, where any camera that agrees on the plane fits equally well. The second model fits the plane of the landmarks by principal component analysis and a homography $\boldsymbol{H}$ from plane coordinates to the image, from four correspondences, lifted to $\boldsymbol{P}_t = \boldsymbol{H}\boldsymbol{F}$ with $\boldsymbol{F}$ the projection onto the plane. The plane model is preferred unless the full camera explains at least a tenth more correspondences. When no camera is accepted, MapSORT falls back to the Mahalanobis gate (Sect. 3.7).

Each track also learns where its landmark projects within its box. When a track is matched under an accepted camera, the residual $\boldsymbol{r} = (\boldsymbol{c}_t^i - \pi(\boldsymbol{P}_t\hat{\boldsymbol{\mu}}_k))/s_t^i$ updates an exponential moving average $\boldsymbol{o}_k$ (rate 0.3) of this offset, and a second average ρ_k of $\|\boldsymbol{r} - \boldsymbol{o}_k\|$ measures how steadily the camera places the track. A track is *reliable* once it has two offsets and $\rho_k \leq 0.3$ box sizes. Its predicted box centre is $\pi(\boldsymbol{P}_t\hat{\boldsymbol{\mu}}_k) + s_k\boldsymbol{o}_k$, with s_k its size.

3.7 Association

Each frame runs three stages.

BoT-SORT association.

Tracks are predicted by BoT-SORT's Kalman filter after compensating global camera motion, and matched to detections on fused IoU and appearance costs (Aharon et al, 2022). These pairs, with the landmark pairs of Sect. 3.6, yield the camera $\boldsymbol{P}_t$.

Map motion.

Reliable tracks that the first stage left unmatched are moved to their predicted position under $\boldsymbol{P}_t$, and the same costs are solved again between them and the detections still unmatched. This recovers tracks whose image-plane motion the Kalman filter could not follow, without disturbing any pair the first stage made. A track still unmatched keeps BoT-SORT's prediction.

Revival.

Detections still unmatched are offered to lost tracks, those missed for at most $A_{\max}$ frames, and to dormant tracks (Sect. 3.8), before any new identity is issued. Each candidate's box is predicted at its projected position and its last size not cut by the frame edge, and clipped to the frame, as the detector would see it. The revival cost is the distance between the centres of the clipped prediction and the detection, in box sizes of the candidate, and the assignment is solved by the Hungarian method with a gate of one box size. Candidates must have two offsets, so that their projection has been checked against their box. An object re-entering the view shows only a sliver at the frame edge, and the clipped predictions of several candidates crowd together there. A detection touching the frame edge is therefore only revived by a candidate whose clipped box overlaps it with an IoU of at least 0.3 and is at least half inside the frame. A new track born on such a detection, where a candidate is expected, stays unconfirmed and unreported for up to 8 frames, until enough of the object is visible to revive it or the wait runs out. When no camera is accepted, lost tracks are revived on the Mahalanobis distance of their latest observation and dormant tracks on that of their fused landmark.

Detections that remain unmatched start tentative tracks, which BoT-SORT confirms on a second match. Every matched or revived track fuses the detection's landmark observation by Eq. (4).

Algorithm 1 MapSORT update at frame t

Require: detections D_t with embeddings f_t^i and landmark observations z_t^i (possibly $\varnothing$); active tracks $\mathcal{A}$ (tracked and lost); dormant pool $\mathcal{Z}$

Predict and motion-compensate every $k \in \mathcal{A}$; $(M_1, U_\mathcal{A}, U_D) \leftarrow$ BoT-SORT association
$\boldsymbol{P}_t \leftarrow$ RANSAC resection on M_1 and mutual-nearest landmark pairs within τ
if $\boldsymbol{P}_t$ accepted **then**
    $(M_2, U_\mathcal{A}, U_D) \leftarrow$ re-solve reliable $k \in U_\mathcal{A}$ at $\pi(\boldsymbol{P}_t \hat{\boldsymbol{\mu}}_k) + s_k \boldsymbol{o}_k$ against $i \in U_D$
    $(M_3, U_D) \leftarrow$ Hungarian on clipped reprojection error, lost $k \in U_\mathcal{A}$ and $k \in \mathcal{Z}$, $i \in U_D$, gate 1 box
else
    $(M_3, U_D) \leftarrow$ Hungarian on d^2_{ki}, lost $k \in U_\mathcal{A}$ and $k \in \mathcal{Z}$, $i \in U_D$
end if
for $(k, i) \in M_1 \cup M_2 \cup M_3$ **do**
    Update k with d_t^i (re-initialise if revived); fuse z_t^i into $(\boldsymbol{\Lambda}_k, \boldsymbol{\eta}_k)$
end for
Learn offsets $\boldsymbol{o}_k, \rho_k$ of matched tracks under $\boldsymbol{P}_t$
for every $i \in U_D$ **do**
    Match to a tentative track or start one; at the frame edge, near an expected revival, keep it unconfirmed
end for
Move confirmed tracks lost for more than $A_{\max}$ frames with a landmark from $\mathcal{A}$ to $\mathcal{Z}$

3.8 Dormant tracks and identity revival

A frame-to-frame tracker deletes a track after $A_{\max}$ frames without a match, because beyond that horizon its image-plane prediction is no longer informative. For a stationary object, the landmark remains informative indefinitely. MapSORT therefore moves a confirmed track whose landmark has been observed to a dormant pool when it exceeds $A_{\max}$, instead of deleting it. A dormant track keeps its identity, fused landmark and offset. When the revival stage matches it to a detection, its Kalman state is re-initialised from the detection box, since the image-plane motion before the absence says nothing about where the object reappears, and its identity and landmark carry over unchanged. The dormant pool is unbounded, so an object keeps its identity across absences of any length.

3.9 Datasets

MapSORT is evaluated on four publicly available agricultural MOT datasets (Table 1, Fig. 3). Together they cover four crops, four acquisition platforms and three viewing patterns: repeated passes along a row, multi-perspective inspection and continuous sweeps. Mask annotations are converted to axis-aligned boxes in MOT format. Per-sequence statistics are listed in Appendix A.

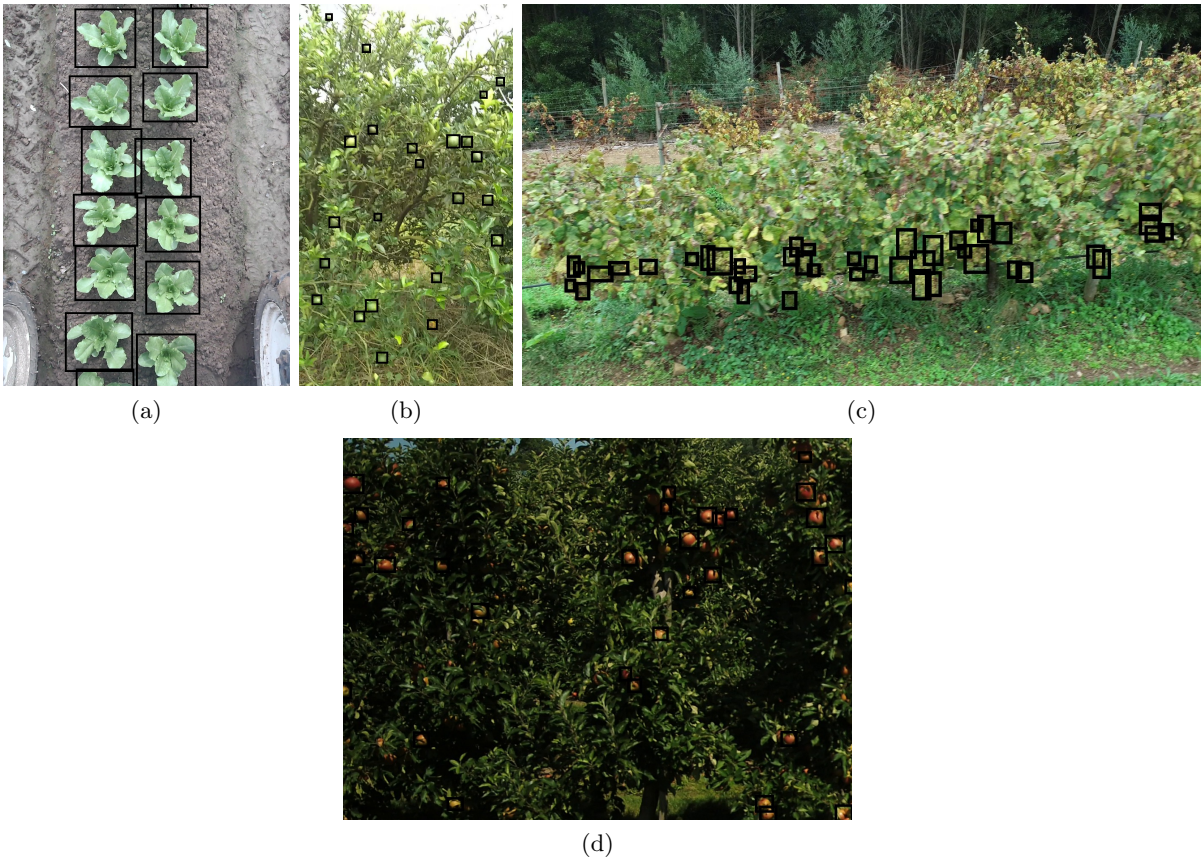
LettuceMOTS (Hu et al, 2024) was recorded by a downward-facing camera on a field robot over lettuce at three growth stages. The robot travels back and forth along the rows, so every plant is seen twice, with long absences in between. The three designated test sequences (0003, 0007, 0011) are used.
MOrangeT (Santos et al, 2024) consists of smartphone videos, each sweeping one side of a sweet-orange tree in a commercial grove; the same fruit leaves and re-enters the field of view several times as the camera moves around the canopy. V01, V02 and V05 to V12 are used.
GrapeMOTS (Ariza-Sentís et al, 2024) was recorded by a UAV over a vineyard. Its path-planned videos view the canopy from multiple perspectives to reduce leaf occlusion. PathPlanning_1, PathPlanning_2, PathPlanning_3 and PathPlanning_7 are used.
AppleMOTS (de Jong et al, 2022) contains UAV and wearable-camera video of apple trees with dense clusters of small, similar fruit. The test split (0006, 0007, 0008, 0010) is used; sequences 0011 and 0012 repeat 0007 and 0008 with locally edited pixels and are excluded. AppleMOTS sequences are continuous sweeps without revisits, and test MapSORT in short-term association among dense, near-identical fruit.

3.10 Detections

Every tracker receives the same detections, produced by a YOLO detector (Jocher et al, 2023) fine-tuned separately for each dataset. On the three MOTS datasets it is the instance-segmentation variant, whose masks the landmark filter of Sect. 3.4 uses. The detector is YOLOv8m (YOLOv8m-seg on the MOTS datasets) at an input size of 1280 px, initialised from COCO weights and fine-tuned on the same sequences

Table 1 Evaluation datasets. Frames, identities, boxes and revisit events are totals over the sequences used. A revisit event is a ground-truth object reappearing after at least 10 frames out of view.

Dataset	Crop	Platform, view	Seqs	Frames	Resolution	IDs	Boxes	Revisits
LettuceMOTS	Lettuce	Robot, nadir	3	712	810×1080	114	9 718	70
MOrangeT	Orange	Phone, lateral	10	3 695	≈1080×1920	1 003	102 409	950
GrapeMOTS	Grape	UAV, multi-view	4	1 761	≤3840×2160	196	25 093	23
AppleMOTS	Apple	UAV, wearable	4	626	1296×972	977	27 533	0



(a) (b) (c) (d)

Fig. 3 Sample frames: (a) LettuceMOTS 0011, (b) MOrangeT V01, (c) GrapeMOTS PathPlanning_7 and (d) AppleMOTS 0008, with their ground-truth boxes.

outside the test data that train the re-identification model (Sect. 3.11). Detections with a confidence above 0.1 are passed to the trackers. On the evaluation sequences the detectors reach an AP_{50} of 96.2 on LettuceMOTS, 87.4 on MOrangeT, 89.1 on GrapeMOTS and 81.3 on AppleMOTS.

3.11 Baselines

MapSORT is compared with three trackers that represent the main current approaches to tracking-by-detection. StrongSORT (Du et al, 2023) associates on appearance first, through a matching cascade over per-track feature galleries gated by Kalman motion, and compensates camera motion with ECC image alignment. Deep OC-SORT (Maggiolino et al, 2023) extends observation-centric motion with sparse optical-flow camera-motion compensation and adaptively weighted appearance. BoT-SORT (Aharon et al, 2022) fuses motion and appearance costs after camera-motion compensation. All three are run from BoxMOT (Broström, 2024) with their default configurations and with the same detections as MapSORT. The appearance-based trackers, MapSORT included, share one re-identification model per dataset: OSNet-x0.25 (Zhou et al, 2019), initialised from MSMT17 and fine-tuned on ground-truth object identities from sequences outside the test data: AppleMOTS 0000 to 0004, GrapeMOTS PathPlanning_4 to _6, LettuceMOTS 0000, 0004 and 0008, and MOrangeT V03. One further sequence per dataset (AppleMOTS 0005, GrapeMOTS PathPlanning_8, LettuceMOTS 0009, MOrangeT V04) selects the training epoch. Each dataset contributes at most 200 identities with up to 10 crops each, spread evenly along each track, and

the model is trained for 10 epochs with label-smoothed cross-entropy and a batch-hard triplet loss. BoT-SORT is also MapSORT's base tracker and shares all of its settings, so comparing the two isolates the effect of the camera and landmark stages.

3.12 Evaluation metrics

Tracking accuracy is measured with HOTA and its detection and association components DetA and AssA (Luiten et al, 2021), MOTA and identity switches (IDSW) from the CLEAR metrics (Bernardin and Stiefelhagen, 2008), and IDF1 (Ristani et al, 2016), all computed with TrackEval (Luiten and Hoffhues, 2020) at an IoU threshold of 0.5 for CLEAR and Identity. Two further metrics address the requirements of agricultural use.

Count error.

Yield estimation from video counts tracks. The count error

$$\mathrm{CE} = 100 \cdot \frac{|\mathrm{IDs}_{\mathrm{pred}}| - |\mathrm{IDs}_{\mathrm{GT}}|}{|\mathrm{IDs}_{\mathrm{GT}}|} \tag{6}$$

is the relative over-count (positive) or under-count (negative) of distinct identities; every identity switch that issues a new identity adds to it.

Revisit identity preservation.

A revisit event is a ground-truth object that reappears after at least $G = 10$ consecutive frames out of view. At each event, the object's tracker identity is read, using exactly the per-frame assignment that CLEAR makes, at the first frame within a grace window of $W = 5$ frames in which it is matched; the grace window allows for track confirmation after reappearance. The event is *preserved* if that identity equals the one the object carried before the absence, a *switch* if it differs, and *unmatched* if the object is not matched within the window. The revisit identity preservation rate is

$$\mathrm{RIP} = 100 \cdot \frac{\#\mathrm{preserved}}{\#\mathrm{preserved} + \#\mathrm{switch}}. \tag{7}$$

Because the per-frame assignment is CLEAR's own, every revisit switch is one of CLEAR's IDSW, so RIP measures the share of the hardest identity decisions, those after an absence, that a tracker gets right. Revisit outcomes are also reported as a function of absence length (Sect. 4.3).

3.13 Implementation details

Table 2 lists MapSORT's settings. They are identical for every sequence and dataset, except for the landmark point filters of Sect. 3.4, which are chosen per sequence; BoT-SORT's own stages use its BoxMOT defaults unchanged. Experiments ran on a laptop with an Intel Core i9-14900HX CPU and an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB). Scene maps were built, and the throughput of Sect. 4.6 measured, on a CPU-only workstation with an Intel Xeon E3-1240 v5 CPU (4 cores, 8 threads, 3.5 GHz).

4 Results

4.1 Comparison with state-of-the-art trackers

Table 3 compares MapSORT with the three baselines on all four datasets. MapSORT achieves the highest HOTA, AssA, MOTA, IDF1 and RIP and the fewest identity switches on every dataset. The gains concentrate in association: DetA, which reflects localisation, is within one point of BoT-SORT's on every dataset, while AssA rises by 18.0 points on average over BoT-SORT, from 31.4 and 36.3 points on LettuceMOTS and MOrangeT to 3.4 and 0.7 points on GrapeMOTS and AppleMOTS. Pooled over the 1043 revisit events of the test data, MapSORT preserves identity in 91.9% of them, against 11.5% for StrongSORT, 9.1% for BoT-SORT and 7.8% for Deep OC-SORT. Identity switches fall from 1157 for BoT-SORT, 1150 for Deep OC-SORT and 2014 for StrongSORT to 223, and the mean absolute count error over the four datasets from 44.9% for the best baseline, Deep OC-SORT, to 5.9%, which translates directly into more accurate yield estimates. Among the baselines, BoT-SORT has the highest HOTA on every dataset. StrongSORT and Deep OC-SORT have lower DetA, 90.9 and 93.2 on average against 97.0

Table 2 MapSORT settings. Box sizes are geometric means of box width and height.

Component	Parameter	Value
Detection	Detector	YOLOv8m (-seg on MOTS datasets), fine-tuned per dataset
	Input size	1280 px
	Confidence threshold	0.1
Scene map	SfM	COLMAP 3.9.1, shared RADIAL camera; MOrangeT: models shipped with the dataset
	Features	SIFT, $\leq$16 384 per frame, max. side 3840 px
	Matching	Exhaustive; sequential, 30 neighbours, on GrapeMOTS PathPlanning_1 and _3
Landmark filters	LettuceMOTS	Mask
	MOrangeT	Track length $\ell_{\min} = 5$
	GrapeMOTS	Mask eroded by 0 to 16 px; depth $k = 1.5$ to 3; $\ell_{\min} = 3$ to 5 on two sequences
	AppleMOTS	Mask eroded by 0 or 1 px; depth $k = 3$ on 0007
BoT-SORT	Re-ID model	OSNet-x0.25, fine-tuned per dataset
	Appearance, proximity thresholds	0.62, 0.61
	First-stage match threshold	0.77
	New-track threshold	0.62
	Lost-track buffer $A_{\max}$	40 frames at 30 fps
Landmarks	Mahalanobis gate τ	200
	Gate for a lost track's latest observation	10
Camera	RANSAC iterations, inlier, min. inliers	300, 0.5 box size, 8
	Plane preferred unless full camera explains more by	10%
Map motion	Offset average rate, min. offsets	0.3, 2
	Reliable residual ρ_k	$\leq$0.3 box size
Revival	Reprojection gate	1.0 box size
	Edge detections: IoU, visible share, wait	0.3, 0.5, 8 frames
	Dormant pool size	Unbounded
Evaluation	IoU threshold	0.5
	Revisit gap G, grace W	10, 5 frames
	Toolkit	TrackEval 1.3.0

Table 3 Tracking results. CE: count error (%). RIP: revisit identity preservation rate (%); AppleMOTS has no revisit events. Best per dataset in bold.

Dataset	Tracker	HOTA↑	AssA↑	DetA↑	MOTA↑	IDF1↑	IDSW↓	CE↓	RIP↑
LettuceMOTS	BoT-SORT	77.80	63.15	**95.83**	95.03	64.93	78	+67.5	1.4
	StrongSORT	70.16	53.25	92.45	90.76	57.74	164	+111.4	2.9
	Deep OC-SORT	77.34	64.50	92.74	92.07	67.46	65	+56.1	6.6
	MapSORT	**95.09**	**94.51**	95.67	**95.62**	**96.53**	**5**	**+3.5**	**97.1**
MOrangeT	BoT-SORT	76.99	60.31	98.27	97.43	66.92	863	+84.6	9.9
	StrongSORT	75.74	59.66	96.14	95.24	66.98	933	+83.8	12.4
	Deep OC-SORT	74.63	59.06	94.32	93.52	66.43	829	+82.3	8.0
	MapSORT	**97.86**	**96.66**	**99.07**	**99.00**	**97.82**	**90**	**+6.0**	**91.8**
GrapeMOTS	BoT-SORT	92.38	86.29	98.90	98.61	89.99	82	+30.1	0.0
	StrongSORT	87.03	78.31	96.72	95.77	84.12	255	+69.4	0.0
	Deep OC-SORT	90.90	85.25	96.92	96.52	88.97	103	+36.7	4.3
	MapSORT	**94.41**	**89.69**	**99.39**	**99.27**	**93.69**	**42**	**-9.2**	**78.3**
AppleMOTS	BoT-SORT	94.06	92.95	95.17	94.68	95.27	134	**+2.0**	n/a
	StrongSORT	78.29	78.30	78.27	75.87	79.15	662	+10.3	n/a
	Deep OC-SORT	88.62	88.26	88.98	88.42	91.88	153	+4.5	n/a
	MapSORT	**94.73**	**93.67**	**95.80**	**95.49**	**95.84**	**86**	-4.8	n/a

for BoT-SORT, because at their default settings a new track is reported only after three consecutive matches, which costs the first frames of every object; this weighs most on AppleMOTS, whose short sequences are dense with briefly visible fruit.

Table B2 in Appendix B reports every test sequence. MapSORT has fewer identity switches than every baseline on 19 of the 21 sequences, and as few as BoT-SORT on the other two: GrapeMOTS PathPlanning_1, where neither makes any, and AppleMOTS 0008, where both make one and Deep OC-SORT none. Its HOTA is the highest of all trackers on 18 sequences, equal to BoT-SORT's on PathPlanning_1, and below the best baseline's on two, GrapeMOTS PathPlanning_3 and AppleMOTS 0007, which Sect. 5 examines.

Table 4 Results on the representative sequences. Columns as in Table 3. AppleMOTS 0008 has no revisit events.

Sequence	Tracker	HOTA↑	AssA↑	DetA↑	MOTA↑	IDF1↑	IDSW↓	CE↓	RIP↑
LettuceMOTS 0011	BoT-SORT	77.88	63.23	**95.93**	95.15	64.91	24	+64.9	0.0
	StrongSORT	76.33	62.17	93.72	92.97	64.03	23	+62.2	0.0
	Deep OC-SORT	77.49	64.93	92.48	91.83	68.03	20	+54.1	5.9
	MapSORT	**93.34**	**91.48**	95.25	**95.18**	**93.67**	**2**	**+5.4**	**95.0**
MOrangeT V01	BoT-SORT	74.39	56.94	97.18	95.85	64.03	105	+93.6	10.0
	StrongSORT	70.32	53.02	93.27	91.60	61.47	131	+109.1	9.6
	Deep OC-SORT	69.40	53.36	90.25	88.97	62.05	101	+91.8	5.8
	MapSORT	**97.07**	**95.75**	**98.42**	**98.23**	**97.08**	**15**	**+8.2**	**90.3**
GrapeMOTS PathPlanning_7	BoT-SORT	89.60	80.92	99.21	98.86	85.48	34	+43.1	0.0
	StrongSORT	77.59	62.56	96.23	94.94	71.78	123	+127.7	0.0
	Deep OC-SORT	90.07	83.02	97.71	97.30	86.71	37	+50.8	4.3
	MapSORT	**96.73**	**93.89**	**99.66**	**99.59**	**96.35**	**10**	**0.0**	**78.3**
AppleMOTS 0008	BoT-SORT	98.79	98.80	98.78	98.77	99.37	1	**0.0**	n/a
	StrongSORT	96.76	96.73	96.80	96.77	98.29	3	-0.6	n/a
	Deep OC-SORT	96.63	96.78	96.49	96.49	98.21	**0**	-1.8	n/a
	MapSORT	**98.81**	**98.82**	**98.79**	**98.79**	**99.38**	1	**0.0**	n/a

4.2 Representative sequences

One representative sequence per dataset is used for the detailed analyses that follow: LettuceMOTS 0011, MOrangeT V01, GrapeMOTS PathPlanning_7 and AppleMOTS 0008. Table 4 reports all trackers on these sequences.

On LettuceMOTS 0011 the robot drives out along two rows and returns, so each of the 37 plants is seen twice. BoT-SORT gives every returning plant a new identity, preserving none of the 22 revisits, and over-counts the plants by 64.9%; StrongSORT also preserves none, and Deep OC-SORT 5.9%. MapSORT preserves 95.0% of the revisits with two identity switches in the whole sequence, raising HOTA from 77.9 to 93.3 and IDF1 from 64.9 to 93.7 (Fig. 1).

On MOrangeT V01, a handheld sweep around an orange tree with 104 revisit events, MapSORT raises HOTA from 74.4 to 97.1 and IDF1 from 64.0 to 97.1, cuts identity switches from 105 to 15, and preserves identity in 90.3% of revisits, against 10.0% for BoT-SORT, the most of any baseline. The count error falls from +93.6% to +8.2%. This holds although the oranges' landmarks rest on few points (a median of two per detection, Table 6): the camera is fitted to many landmarks at once, so a poorly supported landmark still projects close to its object.

On GrapeMOTS PathPlanning_7, where the UAV views the vines from several perspectives, MapSORT raises HOTA from 89.6 to 96.7 and IDF1 from 85.5 to 96.4, cuts identity switches from 34 to 10, and brings the count error from +43.1% to 0.0%. BoT-SORT and StrongSORT preserve none of the 23 revisits and Deep OC-SORT one; MapSORT preserves 78.3% of them. On AppleMOTS 0008, which has no revisits and on which BoT-SORT already makes a single identity switch, MapSORT matches it (HOTA 98.81 against 98.79, one switch each); Deep OC-SORT makes none there, at 96.63 HOTA.

4.3 Identity persistence across revisits

Fig. 4 breaks revisit outcomes down by how long the object was out of view. The baselines preserve identity only for short absences: 57.7% (BoT-SORT), 74.8% (StrongSORT) and 61.7% (Deep OC-SORT) of events with a gap below 30 frames, and at most 3.0% beyond, because a track lost for longer than its buffer is deleted. MapSORT preserves between 85.6% and 93.8% of events in every bin, and its rate does not fall with the length of the absence: it is 93.8% for absences of 120 frames or more. This is the behaviour Sect. 3.8 is designed to produce: the landmark of a stationary object does not age, so a dormant track is as recoverable after a long absence as after a short one.

4.4 Ablation study

Table 5 adds MapSORT's components one at a time to BoT-SORT (a). Keeping lost tracks dormant and reviving them by the Mahalanobis gate on their 3D landmarks (b) raises pooled revisit preservation from 6.9% to 67.3%. Its effect depends on how well the landmarks are supported. On LettuceMOTS 0011, whose landmarks rest on a median of 79 map points, the 3D gate alone reaches the best HOTA of all variants, 97.0. On MOrangeT V01, with two points per landmark, it reaches 92.2, and on GrapeMOTS PathPlanning_7 93.4. Reviving through the per-frame camera instead (c) raises pooled preservation to 85.1%, and HOTA to 96.1 on V01 and 97.4 on PathPlanning_7, because projection removes the depth error that lets adjacent fruit pass a 3D gate; on the planar lettuce scene it is 5.6 points below the 3D

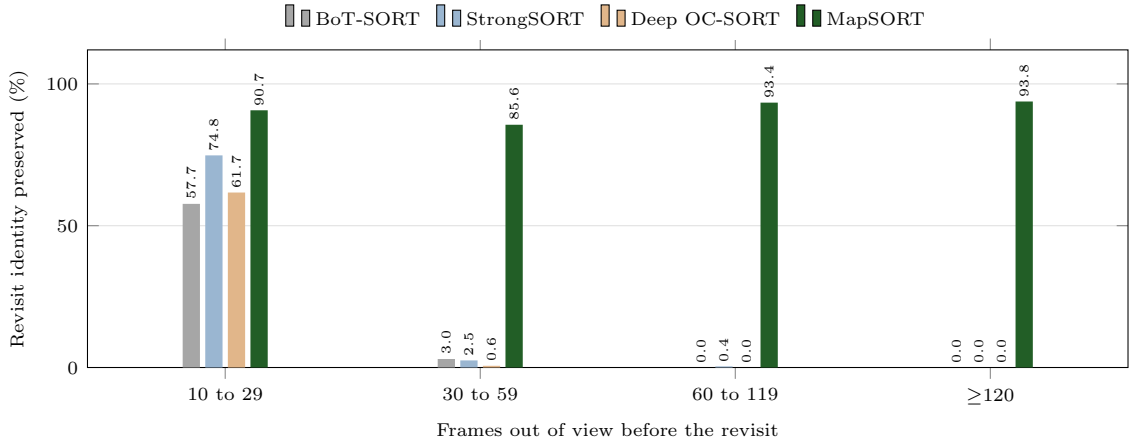


Fig. 4 Revisit identity preservation by length of absence, over the 1043 revisit events of the test sequences. The baselines delete tracks lost for longer than their track buffer and preserve almost no identity beyond that horizon; MapSORT's dormant tracks keep identities recoverable at every absence length.

Table 5 Ablation on the representative sequences (HOTA / IDF1). RIP is pooled over the revisit events of the four sequences (AppleMOTS 0008 has none). (e) uses the nearest SfM keypoint to the box centre as a point landmark with a Euclidean distance gate.

	Lettuce 0011		Orange V01		Grape PP_7		Apple 0008		
Variant	HOTA	IDF1	HOTA	IDF1	HOTA	IDF1	HOTA	IDF1	RIP
(a) BoT-SORT	77.88	64.91	74.39	64.03	89.60	85.48	98.79	99.37	6.9
(b) + landmark revival	**97.00**	**98.46**	92.18	89.57	93.42	92.39	**98.81**	**99.38**	67.3
(c) + camera revival	91.42	90.71	96.14	95.78	**97.41**	**97.51**	**98.81**	**99.38**	85.1
(d) + map motion (MapSORT)	93.34	93.67	97.07	97.08	96.73	96.35	**98.81**	**99.38**	**89.0**
(e) (d), point landmark	94.05	95.21	**97.10**	**97.09**	97.19	97.36	**98.81**	**99.38**	86.9

gate. Map motion, with the learned offsets that revival then also requires of its candidates (d), raises pooled preservation to 89.0% and HOTA to 93.3 on lettuce and 97.1 on V01, at a cost of 0.7 HOTA on PathPlanning_7. On AppleMOTS 0008 every variant stays within 0.02 HOTA of BoT-SORT. Replacing the Gaussian landmark by the nearest triangulated keypoint to the box centre with a Euclidean gate (e), the point-estimate localisation of prior fruit-tracking work, changes HOTA by less than one point on every sequence and lowers pooled preservation to 86.9%: once association runs through the camera, the landmark mostly needs to be a good point, and its covariance matters where the 3D gate is used.

4.5 Sensitivity to the landmark gate

Fig. 5 sweeps the landmark gate τ over two orders of magnitude, from the 95% quantile of the χ^2_3 distribution (7.81) to 1600. In MapSORT, τ gates the landmark pairs that help seed the camera fit and the revival of tracks in frames without a camera. HOTA varies by less than 0.1 points on LettuceMOTS 0011, MOrangeT V01 and AppleMOTS 0008 and revisit preservation not at all. On GrapeMOTS PathPlanning_7, where the camera depends most on the landmark pairs, preservation is 65.2% for $\tau \leq 25$, 73.9% at 50 and 78.3% from 100 upwards. The default $\tau = 200$ lies on that plateau, and a single value serves all datasets, as Proposition 1 predicts.

4.6 Scene maps and runtime

Every frame of every representative sequence is registered into a single model (Table 6), with mean reprojection errors between 0.7 and 1.2 pixels. Building a map on the CPU-only workstation takes 49 minutes for LettuceMOTS 0011, 183 minutes for AppleMOTS 0008 and 352 minutes for the 4K frames of GrapeMOTS PathPlanning_7, of which exhaustive matching takes 69% to 79%. Most detections receive a landmark: 96.7% on LettuceMOTS, 97.9% on AppleMOTS, 93.3% on GrapeMOTS and 76.9% on MOrangeT, where small, glossy fruit seen against foliage yield fewer stable keypoints. Fig. 6 shows the reconstruction of LettuceMOTS 0011: the two rows of plants and the camera path along them, with the outbound and return passes registered into the same map.

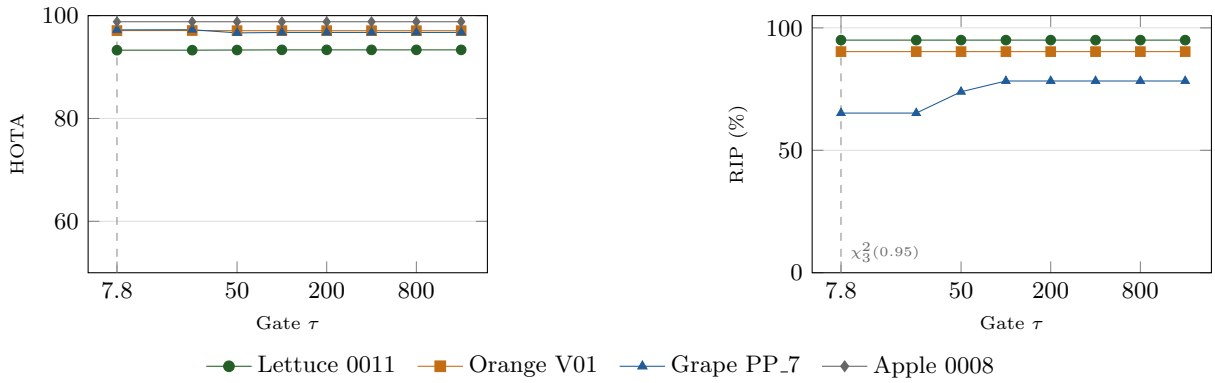


Fig. 5 Sensitivity of HOTA (left) and revisit identity preservation (right) to the landmark gate τ on the representative sequences. The dashed line marks the χ^2_3 95% quantile. AppleMOTS 0008 has no revisit events and appears only on the left.

Table 6 Scene maps on the representative sequences. Track len.: mean number of frames observing a 3D point. Cov.: share of detections that receive a landmark. Pts/LM: median number of map points inside a detection with a landmark. SfM: wall-clock time of feature extraction, matching, mapping and the final bundle adjustment on the CPU-only workstation of Sect. 3.13; the MOrangeT model ships with the dataset.

Sequence	Reg. frames	3D points	Track len.	Reproj. (px)	Cov. (%)	Pts/LM	SfM (min)
LettuceMOTS 0011	223/223	33 766	22.7	0.98	96.7	79	49
MOrangeT V01	408/408	69 081	10.0	1.15	76.9	2	n/a
GrapeMOTS PP_7	225/225	129 210	19.1	0.95	93.3	13	352
AppleMOTS 0008	322/322	116 013	15.8	0.71	97.9	5	183

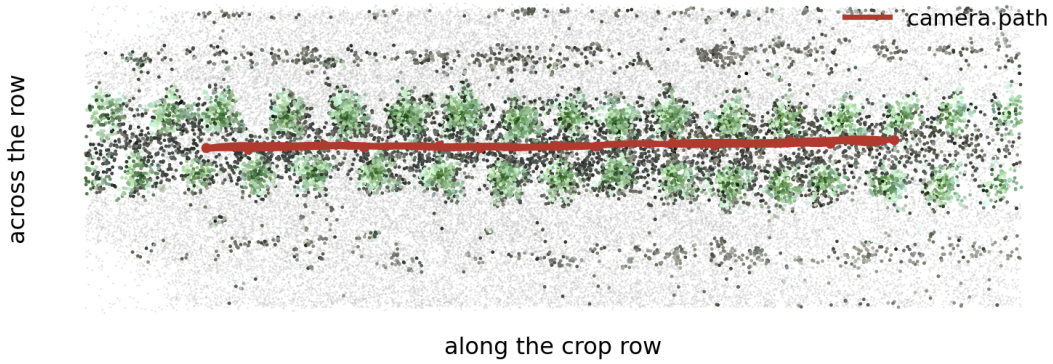


Fig. 6 Sparse scene map of LettuceMOTS 0011, viewed from above. Plant points are green, soil grey; the line is the reconstructed camera path. The outbound and return passes are registered into one map, so each plant has one position on both.

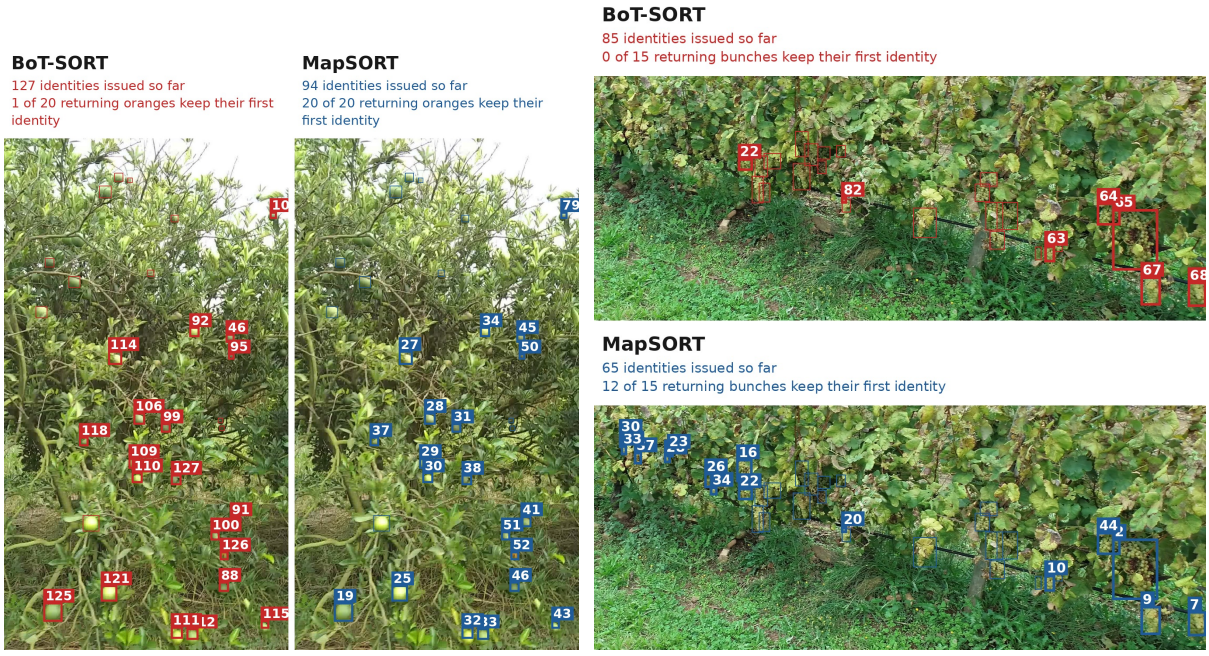
Table 7 reports tracking throughput on the CPU-only workstation. MapSORT runs at 29% to 56% of BoT-SORT's throughput and at 47% to 70% of StrongSORT's, the slowest baseline; Deep OC-SORT runs within 3% of BoT-SORT. Most of the difference is the camera fit, two RANSAC estimates per frame, each with both camera models; landmark fusion operates on 3×3 matrices and costs little. The scene map is built once per sequence, before tracking.

4.7 Qualitative results

Fig. 1 compares BoT-SORT and MapSORT on the return pass of LettuceMOTS 0011. By this frame BoT-SORT has issued 56 identities for 37 plants: every plant on screen carries a new identity, assigned when it reappeared. MapSORT shows each plant with the identity it received on the outbound pass and has issued 39 identities. Fig. 7 shows the same comparison on MOrangeT V01, 232 frames into the handheld sweep, and on GrapeMOTS PathPlanning_7 at the end of the flight. Of the 20 oranges in view that have returned after at least 10 frames out of view, BoT-SORT still gives one its first identity and MapSORT all 20; of the 15 returning bunches, BoT-SORT gives none its first identity and MapSORT 12. By these frames BoT-SORT has issued 127 identities against MapSORT's 94 on V01, and 85 against 65

Table 7 Tracking throughput (frames per second, including image decoding, detection and appearance embedding; excluding map construction), on the CPU-only workstation of Sect. 3.13 (W).

Sequence	Hardware	BoT-SORT	StrongSORT	Deep OC-SORT	MapSORT
LettuceMOTS 0011	W	5.36	3.26	5.36	1.53
MOrangeT V01	W	3.84	2.66	3.82	1.37
GrapeMOTS PP_7	W	1.47	1.18	1.47	0.83
AppleMOTS 0008	W	2.56	1.93	2.50	1.09



(a) (b)

Fig. 7 BoT-SORT and MapSORT on (a) MOrangeT V01, frame 232, and (b) GrapeMOTS PathPlanning_7, frame 221, cropped to the vine. Every box a tracker reports is drawn. The objects that have returned after at least 10 frames out of view are drawn thicker and tagged with their identity, numbered in order of first appearance; under MapSORT they keep their first identity, under BoT-SORT almost none does.

on PathPlanning_7, whose 65 bunches MapSORT counts exactly. AppleMOTS has no revisits to show in this way; Sect. 5 examines it, with the other datasets, through the whole of each sequence.

5 Discussion

The results show that, for stationary agricultural targets, a scene-map position is the identity cue that survives the conditions under which appearance and image-plane motion fail. The effect is largest exactly where agricultural tracking is hardest. When the camera returns to an object after a long absence, BoT-SORT's identity is lost by construction, and MapSORT recovers it at a rate that does not decay with the length of the absence (Fig. 4). Fig. 8 shows the same thing object by object on the four representative sequences: each row is a ground-truth object, coloured by whether the tracker still gives it the identity it first received. The datasets exercise the cue in different ways, and the camera that turns it into a prediction behaves differently on each.

The per-frame camera.

Everything MapSORT adds rests on the camera fitted in each frame, and that camera is almost always available: one is accepted in 99.0% to 99.8% of the frames of each representative sequence and of AppleMOTS 0007. The tracks matched in a frame agree with its camera to a median of 0.03 to 0.06 box sizes, with 90th percentiles of 0.12 to 0.17, once their learned offset is added. The landmark projection alone leaves median errors of 0.08 to 0.17 box sizes: a landmark is the median of the map points in its box, not the box centre, and the offset, which learns where in the box it sits, cuts the error by a factor of about three. These errors are small against the spacing of the objects, a median of 0.95 to 3.4 box sizes to the nearest neighbour, which is why a revival gate of one box size rarely admits the wrong object. Fig. 9

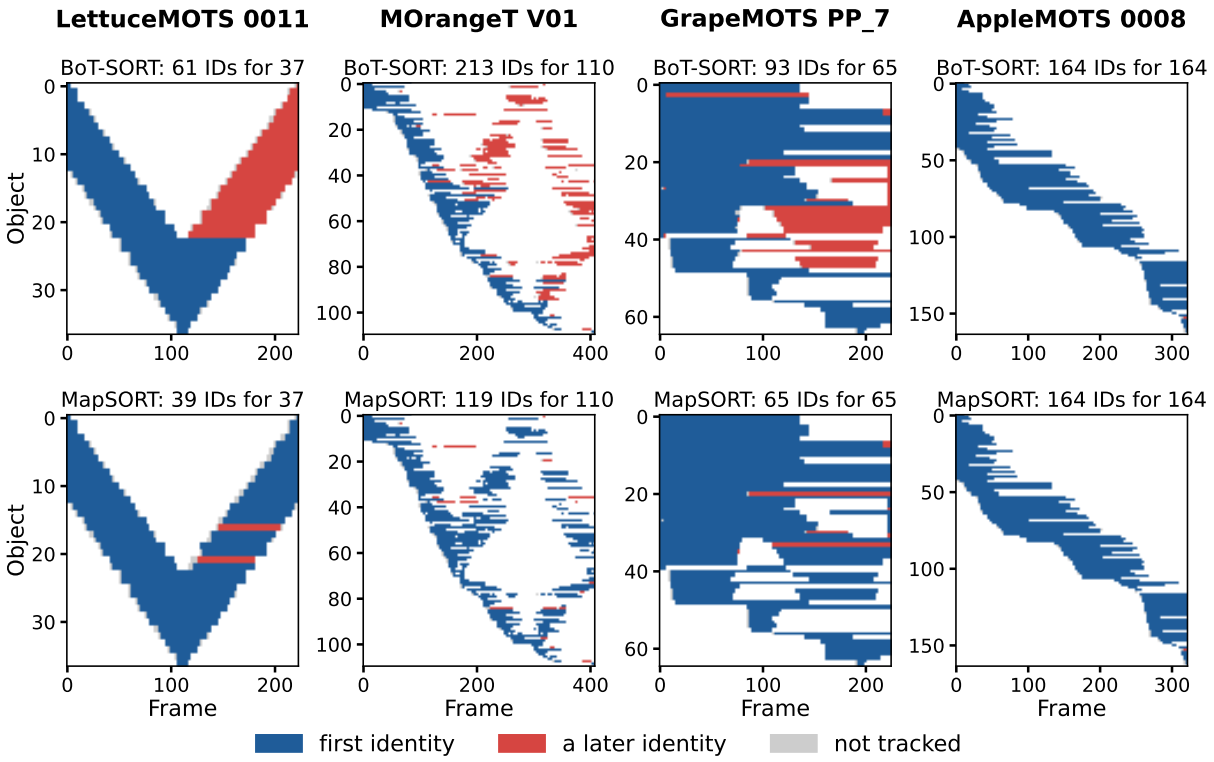


Fig. 8 Identity timelines on the representative sequences. Each row is a ground-truth object, ordered by first appearance, and each column a frame. Blue: the object carries the first identity the tracker gave it; red: a later one; grey: in view but not tracked; white: out of view. Panel titles give the number of identities issued and of objects. BoT-SORT gives the plants of the LettuceMOTS return pass, and many returning oranges and bunches, new identities; MapSORT keeps their first. AppleMOTS 0008 has no revisits, and the two trackers coincide.

shows where the camera places the lost and dormant tracks in one frame of each sequence, including those outside the view.

Repeated passes over a row.

LettuceMOTS is the operating pattern of a field robot working a row and returning, and the clearest case for the cue: on all three sequences MapSORT preserves at least 95% of the returns, with 0, 3 and 2 identity switches against BoT-SORT's 28, 26 and 24, where no baseline makes fewer than 20, and it counts the 40 plants of 0003 exactly. On the return pass of 0011 BoT-SORT gives every plant a new identity; MapSORT changes the identity of two (Fig. 8). Two parts of the method matter here. The field is close to a plane, on which a full projective camera is ill-conditioned; fitting the plane's homography instead keeps the camera stable. And a returning plant enters the image at its edge, where the detector sees only a sliver of it for the first frames. The camera places the plants of the outbound pass beyond that edge before the robot reaches them (Fig. 9a); comparing clipped predictions and holding a new track at the edge until the plant can be revived is what lets it reclaim its identity before a new one is issued. The ablation also shows that the dense map support of a lettuce field makes the 3D landmark gate alone very effective there, more than the camera-based revival; MapSORT uses the 3D gate only as a fallback, in favour of a single configuration that also serves the sparse orange landmarks.

Sweeps around a canopy.

MOrangeT is the largest and most revisit-heavy dataset, with 950 revisits over ten sequences. A handheld sweep moves the camera close to a three-dimensional canopy, so image-plane motion is dominated by parallax that a global camera-motion warp cannot follow, and fruit go out of view behind leaves and the trunk for hundreds of frames. MapSORT preserves 91.8% of these revisits and has fewer identity switches than every baseline on every sequence, 90 in total against 829 for the best baseline, Deep OC-SORT. On V01, BoT-SORT issues 213 identities for 110 oranges and MapSORT 119 (Fig. 8). Its landmarks there rest on a median of two map points, so a 3D comparison is unreliable. Projected through a camera fitted to many landmarks at once, they place the matched oranges to a median of 0.05 box sizes, while

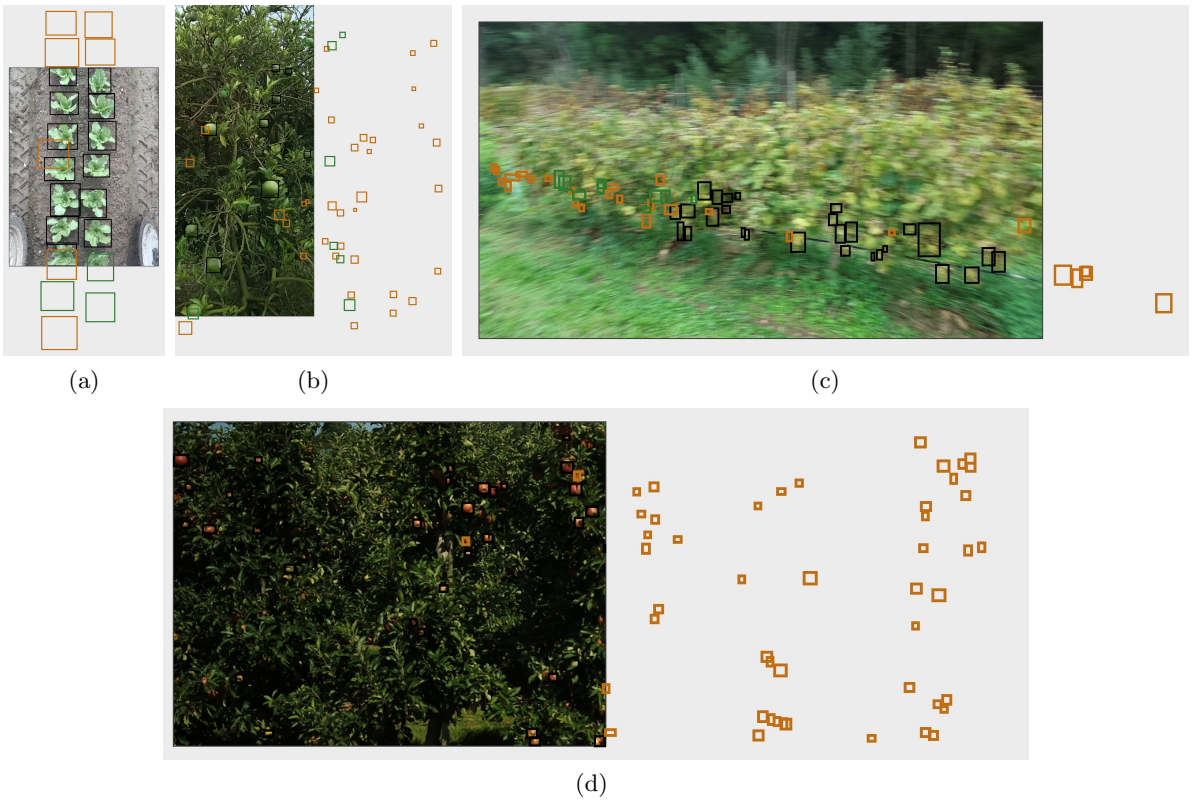


Fig. 9 Where the per-frame camera places the stationary tracks: (a) LettuceMOTS 0011, frame 150, on the return pass; (b) MOrangeT V01, frame 307; (c) GrapeMOTS PathPlanning_7, frame 214; (d) AppleMOTS 0008, frame 150. Black: the boxes of the tracks matched in the frame. Coloured: the predicted boxes of lost and dormant tracks, their landmark's projection plus the learned offset at their last full size; green where the track is matched again within the next 8 frames, orange otherwise. Grey: outside the image. (b) shows a window around the right edge of the frame, and (d) the frame and the area to its right, where the fruit the sweep has passed are placed.

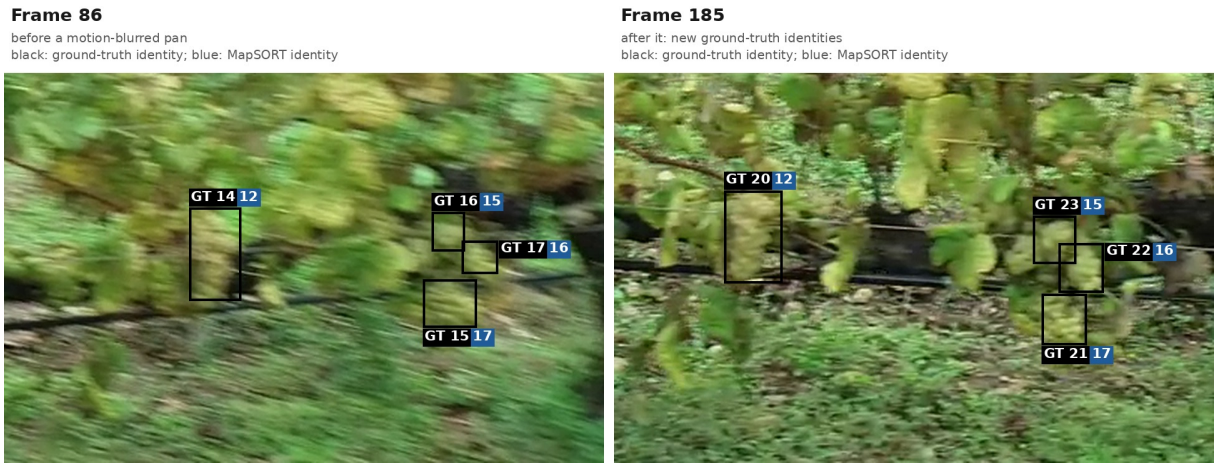
neighbouring oranges are a median of 3.4 box sizes apart. Fruit that leave the view at one edge as the phone sweeps on are placed just beyond it until the sweep brings them back (Fig. 9b).

Multi-perspective UAV inspection.

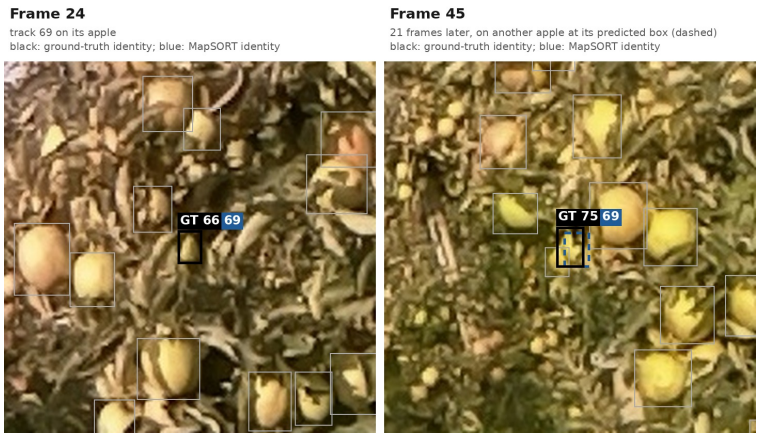
On GrapeMOTS the landmarks of the bunches along a vine lie close to a line, which constrains the camera poorly across it. MapSORT still preserves 78.3% of the revisits of PathPlanning_7, of which the best baseline preserves 4.3%, counts its 65 bunches exactly, and makes no more identity switches than any baseline on every sequence. The camera places the bunches along the vine, including those not detected in the frame (Fig. 9c). On PathPlanning_3, however, its HOTA and IDF1 are below those of all three baselines, and Fig. 10a shows why. A large bunch and three small ones are tracked from the start of the sequence until a fast, motion-blurred pan takes them out of view at frames 88 to 94; they return at frames 156 to 179. The ground truth gives the four bunches new identities, 20 to 23, so the sequence contains no revisit events, while MapSORT gives them back the identities they had. In ground-truth terms each such link merges two objects, which the identity metrics penalise without counting an identity switch. Annotation policies for when an object seen again counts as the same object are not uniform across datasets, and a tracker built to recover identity across absences is, by design, at odds with a policy that resets it.

Dense fruit without revisits.

AppleMOTS has no revisits, and its fruit are small, numerous and alike, so it tests whether the extra stages hurt short-term association. They do not: MapSORT has the highest HOTA on three of four sequences, and fewer identity switches than every baseline on three. Without revisits, the camera only places the apples the sweep has passed beyond the edge of the view (Fig. 9d), where their tracks stay dormant; 111 of the 164 tracks of 0008 end the sequence in the dormant pool. On AppleMOTS 0007 MapSORT is 0.3 HOTA below BoT-SORT although it makes fewer identity switches, 6 against 8. The difference lies in tracks carried onto a different apple: a track whose apple disappears behind foliage is later matched to another apple nearby, which the identity metrics penalise without counting a switch. MapSORT does



(a)



(b)

Fig. 10 Where MapSORT scores below BoT-SORT. (a) GrapeMOTS PathPlanning_3: four bunches before and after a motion-blurred pan. The ground truth gives them new identities when they return; MapSORT gives them back their own, which the identity metrics count as merging two objects. (b) AppleMOTS 0007: a lost track is matched, 21 frames after its apple was last seen, to a different apple 0.25 box sizes from where the camera places it (dashed). Grey: the other annotated apples.

this 16 times on 0007 and BoT-SORT 11 times, in both cases after absences of 1 to 31 frames, within the lost-track buffer. 0007 is where the camera is least precise: the matched apples agree with it to a median of 0.11 box sizes and a 90th percentile of 0.67, against at most 0.17 on the representative sequences, while neighbouring apples are a median of 1.3 box sizes apart. Seven of the 16 transfers land within one box size of where the camera places the lost track, and appearance does not separate the apples. Fig. 10b shows one, after 21 frames, onto an apple 0.25 box sizes from the prediction. The single identity switch on AppleMOTS 0008, which BoT-SORT also makes, is an apple seen in two frames, out of view for four and then seen again. Its first track never received a landmark, so the camera cannot place it, and BoT-SORT's motion model does not recover it; Deep OC-SORT avoids it, at 2.2 points lower HOTA.

Consequences for precision agriculture.

The most direct consequence is counting. Every identity switch that issues a new identity is a phantom plant or fruit in the count. The baselines over-count by 43% to 128% on the representative sequences with revisits, which would make their output unusable as a yield estimate without heavy post-hoc correction; MapSORT's count error there is between 0% and 8%. A second consequence is per-plant records. Because a plant keeps its identity across passes, measurements taken on the outbound and return passes, or from different UAV perspectives, attach to the same plant, which is what plant-level phenotyping and selective treatment need. A third is spatial: every MapSORT track carries a position in the scene map, so a track list is also a plant map. Registering the map to a GNSS-referenced frame, as is routine in orchard and field mapping, would turn MapSORT's output into a georeferenced inventory of individual plants or fruit. MapSORT also composes with an existing tracker rather than replacing it: BoT-SORT's association is

unchanged and runs first, and the camera stages act only on what it leaves unmatched, so improvements in detectors, appearance models or the base tracker carry over directly.

Limitations.

The camera and revival stages cost throughput: on a CPU, MapSORT runs at 29% to 56% of BoT-SORT's frame rate, and building a scene map on a CPU takes from under an hour to several hours. MapSORT's settings are shared across all datasets, but were chosen on the same public sequences on which they are evaluated, since these datasets provide no separate validation split, and the landmark point filters were chosen per sequence on that sequence's own points. Detections without a landmark, such as small fruit at the limit of the map's resolution, benefit only from BoT-SORT's own association, as the switch on AppleMOTS 0008 shows. Among dense, near-identical fruit, a lost track can be matched to a neighbour of its own fruit, as on AppleMOTS 0007. And the cue assumes that the objects do not move: fruit swaying on a branch, or a plant pushed aside by the robot, would carry a landmark that no longer marks where it is.

6 Conclusion

MapSORT turns the stationarity of crops and fruit into an identity cue. It gives every detection and track a Gaussian landmark in a structure-from-motion map of the scene, fits a camera to its own tracks in every frame, and uses that camera to re-match the tracks its base tracker misses and to revive tracks, kept dormant rather than deleted, when their objects return, including at the edge of the image. On four public agricultural datasets covering lettuce, oranges, grapes and apples, recorded from a field robot, a handheld camera, a wearable camera and a UAV, MapSORT outperforms StrongSORT, Deep OC-SORT and BoT-SORT, has fewer identity switches than every baseline on 19 of the 21 test sequences, preserves identity across revisits of any length, and produces substantially more accurate counts. MapSORT currently builds its scene map offline. Recent online reconstruction models that recover scene geometry frame by frame from monocular video (Wang et al, 2025; Murai et al, 2025) are a natural drop-in source for the same landmark observations, and extending MapSORT to online operation on a field robot is the subject of future work.

Statements and Declarations

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Ethics approval and consent. Not applicable.

Data availability. All datasets used are publicly available from their original publishers: LettuceMOTS (Hu et al, 2024), MOrangeT (Santos et al, 2024), GrapeMOTS (Ariza-Sentís et al, 2024) and AppleMOTS (de Jong et al, 2022).

Code availability. The MapSORT source code is available at https://github.com/shinobi-wolf/MapSORT.

Author contributions. Srihari Vemuru conceived the method, implemented it, conducted the experiments and wrote the manuscript.

Appendix A Per-sequence statistics

Table A1 Statistics of every evaluation sequence. Revisits: ground-truth objects reappearing after at least 10 frames out of view.

Dataset	Sequence	Frames	Resolution	GT IDs	Boxes	Revisits
LettuceMOTS	0003	245	810×1080	40	3 304	25
	0007	244	810×1080	37	3 341	23
	0011	223	810×1080	37	3 073	22
MOrangeT	V01	408	1083×1944	110	7 848	104
	V02	409	1063×1890	46	4 354	35
	V05	293	≤1081×1941	148	11 944	125
	V06	447	1074×1910	122	12 411	109
	V07	553	1079×1924	154	19 069	170
	V08	544	1078×1918	192	24 500	209
	V09	222	1084×1951	16	1 923	17
	V10	244	1080×1926	13	1 372	19
	V11	268	1079×1920	87	7 265	64
	V12	307	1079×1920	115	11 723	98
GrapeMOTS	PathPlanning_1	927	1920×1080	29	4 796	0
	PathPlanning_2	209	3840×2160	45	4 838	0
	PathPlanning_3	400	1920×1080	57	6 445	0
	PathPlanning_7	225	3840×2160	65	9 014	23
AppleMOTS	0006	101	1296×972	279	4 524	0
	0007	103	1296×972	255	7 917	0
	0008	322	1296×972	164	10 618	0
	0010	100	1296×972	279	4 474	0

Appendix B Per-sequence results

Table B2 reports HOTA, IDF1 and identity switches of every tracker on each of the 21 test sequences.

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Table B2 HOTA, IDF1 and identity switches of every tracker on every test sequence. Best per sequence in bold.

		BoT-SORT			StrongSORT			Deep OC-SORT			MapSORT		
Dataset	Sequence	HOTA	IDF1	IDSW	HOTA	IDF1	IDSW	HOTA	IDF1	IDSW	HOTA	IDF1	IDSW
LettuceMOTS	0003	77.58	64.69	28	57.59	46.90	117	77.33	67.63	23	**96.40**	**98.15**	**0**
	0007	77.93	65.18	26	75.36	62.47	24	77.20	66.78	22	**95.37**	**97.56**	**3**
	0011	77.88	64.91	24	76.33	64.03	23	77.49	68.03	20	**93.34**	**93.67**	**2**
MOrangeT	V01	74.39	64.03	105	70.32	61.47	131	69.40	62.05	101	**97.07**	**97.08**	**15**
	V02	81.02	71.85	33	78.31	71.32	60	78.07	70.85	29	**94.04**	**93.70**	**14**
	V05	76.03	63.26	111	75.19	64.02	105	73.88	63.42	102	**97.50**	**97.13**	**12**
	V06	75.38	64.46	106	73.38	63.84	126	72.27	63.09	105	**98.56**	**98.73**	**4**
	V07	76.79	66.43	141	74.67	65.09	155	74.41	65.81	137	**98.40**	**98.51**	**12**
	V08	76.50	65.97	185	75.75	66.19	188	74.70	65.69	182	**98.19**	**98.27**	**14**
	V09	82.27	77.16	13	82.60	80.11	12	83.12	79.87	12	**97.85**	**97.50**	**1**
	V10	69.71	59.30	16	82.32	77.09	12	76.50	70.03	13	**93.43**	**90.24**	**4**
	V11	80.21	73.56	61	80.24	75.36	57	77.58	72.63	60	**98.53**	**98.81**	**4**
	V12	79.01	71.24	92	78.07	71.55	87	76.59	70.33	88	**97.93**	**97.85**	**10**
GrapeMOTS	PathPlanning_1	**99.38**	**99.69**	**0**	98.69	99.34	**0**	97.28	97.29	1	**99.38**	**99.69**	**0**
	PathPlanning_2	87.20	83.78	30	81.19	78.37	102	81.04	77.22	43	**89.83**	**89.82**	**21**
	PathPlanning_3	**94.56**	93.73	18	94.22	94.17	30	94.08	**94.67**	22	90.63	88.40	**11**
	PathPlanning_7	89.60	85.48	34	77.59	71.78	123	90.07	86.71	37	**96.73**	**96.35**	**10**
AppleMOTS	0006	84.45	86.09	89	40.66	42.69	263	74.05	79.73	93	**87.89**	**89.43**	**53**
	0007	**95.19**	**96.38**	8	83.32	85.11	128	90.69	93.95	7	94.86	95.80	**6**
	0008	98.79	99.37	1	96.76	98.29	3	96.63	98.21	**0**	**98.81**	**99.38**	1
	0010	89.64	92.28	36	39.21	40.23	268	77.91	83.63	53	**91.21**	**93.66**	**26**

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